Management Opportunities, Challenges and Uncertainties and 2025 Outlook

Automotive—Production

The following is a summary of the status of production of each of our announced vehicle models in production and under development, as of the date of this Annual Report on Form 10-K:

Region	Vehicle Model(s)	Production Status
California	Model S / Model X	Active
	Model 3 / Model Y	Active
Shanghai	Model 3 / Model Y	Active
Berlin	Model Y	Active
Texas	Model Y	Active
	Cybertruck	Active
	Cybercab	In development
Nevada	Tesla Semi	Pilot production
TBD	Roadster	In development

We are focused on growing our manufacturing capacity, which includes capacity for manufacturing newer vehicle models such as our Cybertruck, Tesla Semi and future vehicles utilizing aspects of our next generation platform, and ramping the production at our Gigafactories to their installed production capacities as well as increasing production rate and efficiency at our current factories. The next phase of production growth will depend on the continued ramp at our factories and be initiated by advances in autonomy and the introduction of new products, including those built on our next generation vehicle platform, as well as our ability to add to our available sources of battery cell supply by manufacturing our own cells that we are developing to have high-volume output, lower capital and production costs and longer range. Our goals are to improve vehicle performance, decrease production costs and increase affordability and customer awareness.

These plans are subject to uncertainties inherent in establishing and ramping manufacturing operations, which may be exacerbated by new product and manufacturing technologies we introduce, the number of concurrent international projects, any industry-wide component constraints, labor shortages and any future impact from events outside of our control. For example, during the first quarter of 2024, we experienced a sequential decline in production volumes partially caused by the early phase of the production ramp of the updated Model 3 at our Fremont factory, and factory shutdowns at Gigafactory Berlin-Brandenburg resulting from shipping diversions caused by the Red Sea conflict and an arson attack. In the first quarter of 2025, as we launch our New Model Y worldwide, we may similarly experience delays or declines in production volumes due to simultaneous manufacturing ramps in facilities on three continents. Moreover, we have set ambitious technological targets with our plans for battery cells as well as for iterative manufacturing and design improvements for our vehicles with each new factory.

Automotive—Demand, Sales, Deliveries and Infrastructure

Our cost reduction efforts, cost innovation strategies, and additional localized procurement and manufacturing are key to our vehicles' affordability and have allowed us to competitively price our vehicles. We will also continue to generate demand by improving our vehicles' performance and functionality, including through product offerings and features based on artificial intelligence such as Autopilot, FSD (Supervised), and other software, and delivering new vehicles and vehicle options, such as our launch of the updated Model 3 in 2024, and the New Model Y in the first quarter of 2025. In addition, we have been increasing awareness, and expanding our vehicle financing programs, including attractive leasing terms for our customers.

Furthermore, significant judgment is required in evaluating our tax positions. In the ordinary course of business, there are many transactions and calculations for which the ultimate tax settlement is uncertain. As a result, we recognize the effect of this uncertainty on our tax attributes or taxes payable based on our estimates of the eventual outcome. These effects are recognized when, despite our belief that our tax return positions are supportable, we believe that it is more likely than not that some of those positions may not be fully sustained upon review by tax authorities. We are required to file income tax returns in the U.S. and various foreign jurisdictions, which requires us to interpret the applicable tax laws and regulations in effect in such jurisdictions. Such returns are subject to audit by the various federal, state and foreign taxing authorities, who may disagree with respect to our tax positions. We believe that our consideration is adequate for all open audit years based on our assessment of many factors, including past experience and interpretations of tax law. We review and update our estimates in light of changing facts and circumstances, such as the closing of a tax audit, the lapse of a statute of limitations or a change in estimate. To the extent that the final tax outcome of these matters differs from our expectations, such differences may impact income tax expense in the period in which such determination is made.

Results of Operations

Revenues

	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
(Dollars in millions)	2024	2023	2022	$	%	$	%
Automotive sales	$ 72,480	$ 78,509	$ 67,210	$ (6,029)	(8)%	$ 11,299	17 %
Automotive regulatory credits	2,763	1,790	1,776	973	54 %	14	1 %
Automotive leasing	1,827	2,120	2,476	(293)	(14)%	(356)	(14)%
Total automotive revenues	77,070	82,419	71,462	(5,349)	(6)%	10,957	15 %
Services and other	10,534	8,319	6,091	2,215	27 %	2,228	37 %
Total automotive & services and other segment revenue	87,604	90,738	77,553	(3,134)	(3)%	13,185	17 %
Energy generation and storage segment revenue	10,086	6,035	3,909	4,051	67 %	2,126	54 %
Total revenues	$ 97,690	$ 96,773	$ 81,462	$ 917	1 %	$ 15,311	19 %

Automotive & Services and Other Segment

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, including access to our FSD (Supervised) features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. These deliveries are vehicles that are not subject to lease accounting.

Automotive regulatory credits includes sales of regulatory credits to other automotive manufacturers. Our revenue from automotive regulatory credits is directly related to our new vehicle production, sales and pricing negotiated with our customers. We monetize them proactively as new vehicles are sold based on standing arrangements with buyers of such credits, typically as close as possible to the production and delivery of the vehicle or changes in regulation impacting the credits.

Automotive leasing revenue includes the amortization of revenue for vehicles under direct operating lease agreements. Additionally, automotive leasing revenue includes direct sales-type leasing programs where we recognize all revenue associated with the sales-type lease upon delivery to the customer.

Services and other revenue consists of sales of used vehicles, non-warranty maintenance services and collision, part sales, paid Supercharging, insurance services revenue and retail merchandise sales.

2024 compared to 2023

Automotive sales revenue decreased $6.03 billion, or 8%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to lower average selling price on our vehicles driven by overall price reductions and attractive financing options provided in 2024 as well as mix. Additionally, there was a decrease of approximately 22,000 combined Model 3 and Model Y cash deliveries. The decreases were partially offset by an increase of approximately 19,000 deliveries of other models primarily due to our production ramp of Cybertruck. Additionally, we recognized $596 million of FSD (Supervised) revenue due to release of certain features in 2024.

Automotive regulatory credits revenue increased $973 million, or 54%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, driven by demand for credits in North America as other automobile manufacturers scale back on their battery electric vehicle plans.

Automotive leasing revenue decreased $293 million, or 14%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to lower direct sales-type leasing deliveries as we have shifted towards providing leasing options through commercial banking partner programs that allow for us to recognize upfront revenue in automotive sales and a decrease in lease buyouts.

Services and other revenue increased $2.22 billion, or 27%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to increases in sales of used vehicles, non-warranty maintenance services and collision revenue, paid Supercharging revenue, insurance services revenue and part sales revenue.

Energy Generation and Storage Segment

Energy generation and storage revenue includes sales and leasing of solar energy generation and energy storage products, financing of solar energy generation products, services related to such products and sales of solar energy systems incentives.

2024 compared to 2023

Energy generation and storage revenue increased $4.05 billion, or 67%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to a 16.7 GWh increase in Megapack and Powerwall deployments compared to the prior year.

Cost of Revenues and Gross Margin

	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
(Dollars in millions)	2024	2023	2022	$	%	$	%
Cost of revenues							
Automotive sales	$ 61,870	$ 65,121	$ 49,599	$ (3,251)	(5)%	$ 15,522	31 %
Automotive leasing	1,003	1,268	1,509	(265)	(21)%	(241)	(16)%
Total automotive cost of revenues	62,873	66,389	51,108	(3,516)	(5)%	15,281	30 %
Services and other	9,921	7,830	5,880	2,091	27 %	1,950	33 %
Total automotive & services and other segment cost of revenues	72,794	74,219	56,988	(1,425)	(2)%	17,231	30 %
Energy generation and storage segment	7,446	4,894	3,621	2,552	52 %	1,273	35 %
Total cost of revenues	$ 80,240	$ 79,113	$ 60,609	$ 1,127	1 %	$ 18,504	31 %
Gross profit total automotive	$ 14,197	$ 16,030	$ 20,354				
Gross margin total automotive	18.4 %	19.4 %	28.5 %				
Gross profit total automotive & services and other segment	$ 14,810	$ 16,519	$ 20,565				
Gross margin total automotive & services and other segment	16.9 %	18.2 %	26.5 %				
Gross profit energy generation and storage segment	$ 2,640	$ 1,141	$ 288				
Gross margin energy generation and storage segment	26.2 %	18.9 %	7.4 %				
Total gross profit	$ 17,450	$ 17,660	$ 20,853				
Total gross margin	17.9 %	18.2 %	25.6 %				

Gross margin for energy generation and storage increased from 18.9% to 26.2% in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to margin improvements for our energy storage products driven by cost reductions, including benefits from IRA manufacturing credits, and a higher proportion of our storage business, which operated at a higher gross margin, within the segment as compared to the prior periods.

Research and Development Expense

	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
(Dollars in millions)	2024	2023	2022	$	%	$	%
Research and development	$ 4,540	$ 3,969	$ 3,075	$ 571	14 %	$ 894	29 %
As a percentage of revenues	5 %	4 %	4 %				

Research and development (“R&D”) expenses consist primarily of personnel costs for our teams in engineering and research, manufacturing engineering and manufacturing test organizations, prototyping expense, contract and professional services and amortized equipment expense.

R&D expenses increased $571 million, or 14%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023. The overall increases were primarily driven by additional costs year over year related to AI programs. R&D expenses as a percentage of revenue increased from 4% to 5% in the year ended December 31, 2024 as compared to the year ended December 31, 2023 as we continue to expand our product roadmap and technologies.

Selling, General and Administrative Expense

	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
(Dollars in millions)	2024	2023	2022	$	%	$	%
Selling, general and administrative	$ 5,150	$ 4,800	$ 3,946	$ 350	7 %	$ 854	22 %
As a percentage of revenues	5 %	5 %	5 %				

Selling, general and administrative (“SG&A”) expenses generally consist of personnel and facilities costs related to our stores, marketing, sales, executive, finance, human resources, information technology and legal organizations, as well as fees for professional and contract services and litigation settlements.

SG&A expenses increased $350 million, or 7%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023 driven by a $171 million increase in facilities related expenses, a $115 million increase in employee and labor costs, including professional services and a $57 million increase in marketing expenses.

Restructuring and Other

	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
(Dollars in millions)	2024	2023	2022	$	%	$	%
Restructuring and other	$ 684	$ —	$ 176	$ 684	Not meaningful	$ (176)	(100)%

In the second quarter of 2024, we initiated and substantially completed certain restructuring actions to reduce costs and improve efficiency. As a result, we recognized $583 million of employee termination expenses in Restructuring and other in our consolidated income statement. These expenses were substantially paid with an immaterial accrual remaining in Accrued liabilities and other in our consolidated balance sheet as of December 31, 2024.

Interest Income

	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
(Dollars in millions)	2024	2023	2022	$	%	$	%
Interest income	$ 1,569	$ 1,066	$ 297	$ 503	47 %	$ 769	259 %

Interest income increased $503 million, or 47%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to higher interest earned on our cash and cash equivalents and short-term investments compared to the prior year due to an increase in our portfolio balance and a higher weighted average interest rate.

Other Income (Expense), Net

(Dollars in millions)	Year Ended December 31, 2024	2023	2022	2024 vs. 2023 Change $	%	2023 vs. 2022 Change $	%
Other income (expense), net	$ 695	$ 172	$ (43)	$ 523	304%	$ 215	Not meaningful

Other income (expense), net, consists of foreign exchange gains and losses related to our foreign currency-denominated monetary assets and liabilities and fair value remeasurements of our digital assets following the adoption of ASU 2023-08 effective January 1, 2024. See Note 2, *Summary of Significant Accounting Policies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Other income (expense), net, changed favorably by $523 million in the year ended December 31, 2024 as compared to the year ended December 31, 2023 primarily due to remeasurement of our bitcoin digital assets to fair value in 2024 (see above), partially offset by unfavorable fluctuations in foreign currency exchange rates on our intercompany balances. As our intercompany balances are significant in nature and we do not typically hedge foreign currency risk, we can experience significant fluctuations in foreign currency exchange rate gains and losses from period to period.

Provision for (Benefit from) Income Taxes

(Dollars in millions)	Year Ended December 31, 2024	2023	2022	2024 vs. 2023 Change $	%	2023 vs. 2022 Change $	%
Provision for (benefit from) income taxes	$ 1,837	$ (5,001)	$ 1,132	$ 6,838	Not meaningful	$ (6,133)	Not meaningful
Effective tax rate	20 %	(50)%	8 %				

Our provision for (benefit from) income taxes changed by $6.84 billion in the year ended December 31, 2024 as compared to the year ended December 31, 2023. Our effective tax rate changed to an expense of 20% in the year ended December 31, 2024 from a benefit of 50% in the year ended December 31, 2023. These changes are primarily due to the impact of releasing the valuation allowance on our U.S. deferred tax assets in the fourth quarter of 2023.

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. In the fourth quarter of 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which was objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that most of our U.S. federal and certain state deferred tax assets are realizable and released $6.54 billion of our valuation allowance. As of December 31, 2024, we continue to conclude that it is more likely than not that our deferred tax assets are realizable, except for our California deferred tax assets, U.S. foreign tax credits and certain foreign operating losses.

See Note 13, *Income Taxes*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Liquidity and Capital Resources

We expect to continue to generate net positive operating cash flow. The cash we generate from our core operations enables us to fund ongoing operations and production, our research and development projects for new products and technologies including our proprietary battery cells, additional manufacturing ramps at existing manufacturing facilities, the construction of future factories, and the continued expansion of our retail and service locations, body shops, Mobile Service fleet, Supercharger, including to support NACS, energy product installation capabilities and autonomy and other artificial intelligence enabled products.

In addition, because a large portion of our future expenditures will be to fund our growth, we expect that if needed we will be able to adjust our capital and operating expenditures by operating segment. For example, if our near-term manufacturing operations decrease in scale or ramp more slowly than expected, including due to global economic or business conditions, we may choose to correspondingly slow the pace of our capital expenditures. Finally, we continually evaluate our cash needs and may decide it is best to raise additional capital or seek alternative financing sources to fund the rapid growth of our business, including through drawdowns on existing or new debt facilities or financing funds. Conversely, we may also from time to time determine that it is in our best interests to voluntarily repay certain indebtedness early.

Accordingly, we believe that our current sources of funds will provide us with adequate liquidity during the 12-month period following December 31, 2024, as well as in the long-term.

See the sections below for more details regarding the material requirements for cash in our business and our sources of liquidity to meet such needs.

Material Cash Requirements

From time to time in the ordinary course of business, we enter into agreements with vendors for the purchase of components and raw materials to be used in the manufacture of our products. However, due to contractual terms, variability in the precise growth curves of our development and production ramps, and opportunities to renegotiate pricing, we generally do not have binding and enforceable purchase orders under such contracts beyond the short-term, and the timing and magnitude of purchase orders beyond such period is difficult to accurately project.

As discussed in and subject to the considerations referenced in Part II, Item 7, Management's Discussion and Analysis of Financial Condition and Results of Operations—Management Opportunities, Challenges and Uncertainties and 2025 Outlook—Cash Flow and Capital Expenditure Trends in this Annual Report on Form 10-K, we currently expect our capital expenditures to support our projects globally to exceed $11.00 billion in 2025 and in each of the following two fiscal years.

As of December 31, 2024, we and our subsidiaries had outstanding $7.91 billion in aggregate principal amount of indebtedness, of which $2.35 billion is current. As of December 31, 2024, our total minimum lease payments was $7.05 billion, of which $1.19 billion is due in the succeeding 12 months. For details regarding our indebtedness and lease obligations, refer to Note 10, *Debt*, and Note 11, *Leases*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

Sources and Conditions of Liquidity

Our sources to fund our material cash requirements are predominantly from our deliveries and servicing of new and used vehicles, sales and installations of our energy storage products, interest income, and proceeds from debt facilities and equity offerings, when applicable.

As of December 31, 2024, we had $16.14 billion and $20.42 billion of cash and cash equivalents and short-term investments, respectively. Balances held in foreign currencies had a U.S. dollar equivalent of $2.90 billion and consisted primarily of euros and Chinese yuan. We had $5.00 billion of unused committed credit amounts as of December 31, 2024. For details regarding our indebtedness, refer to Note 10, *Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

We continue adapting our strategy to meet our liquidity and risk objectives, such as investing in U.S. government securities and other investments, invest in autonomy, do more vertical integration, expand our product roadmap and provide financing options to our customers.

Summary of Cash Flows

	Year Ended December 31,		
(Dollars in millions)	2024	2023	2022
Net cash provided by operating activities	$ 14,923	$ 13,256	$ 14,724
Net cash used in investing activities	$ (18,787)	$ (15,584)	$ (11,973)
Net cash provided by (used in) financing activities	$ 3,853	$ 2,589	$ (3,527)

Cash Flows from Operating Activities

Our cash flows from operating activities are significantly affected by our cash investments to support the growth of our business in areas such as research and development and selling, general and administrative and working capital. Our operating cash inflows include cash from vehicle sales and related servicing, customer lease and financing payments, customer deposits, cash from sales of regulatory credits and energy generation and storage products, and interest income on our cash and investments portfolio. These cash inflows are offset by our payments to suppliers for production materials and parts used in our manufacturing process, operating expenses, operating lease payments and interest payments on our financings.

Tesla, Inc.

Consolidated Balance Sheets
(in millions, except per share data)

	December 31, 2024	December 31, 2023
Assets		
Current assets		
Cash and cash equivalents	$ 16,139	$ 16,398
Short-term investments	20,424	12,696
Accounts receivable, net	4,418	3,508
Inventory	12,017	13,626
Prepaid expenses and other current assets	5,362	3,388
Total current assets	58,360	49,616
Operating lease vehicles, net	5,581	5,989
Solar energy systems, net	4,924	5,229
Property, plant and equipment, net	35,836	29,725
Operating lease right-of-use assets	5,160	4,180
Digital assets, net	1,076	184
Intangible assets, net	150	178
Goodwill	244	253
Deferred tax assets	6,524	6,733
Other non-current assets	4,215	4,531
Total assets	$ 122,070	$ 106,618
Liabilities		
Current liabilities		
Accounts payable	$ 12,474	$ 14,431
Accrued liabilities and other	10,723	9,080
Deferred revenue	3,168	2,864
Current portion of debt and finance leases	2,456	2,373
Total current liabilities	28,821	28,748
Debt and finance leases, net of current portion	5,757	2,857
Deferred revenue, net of current portion	3,317	3,251
Other long-term liabilities	10,495	8,153
Total liabilities	48,390	43,009
Commitments and contingencies (Note 14)		
Redeemable noncontrolling interests in subsidiaries	63	242
Equity		
Stockholders' equity		
Preferred stock; $0.001 par value; 100 shares authorized; no shares issued and outstanding	—	—
Common stock; $0.001 par value; 6,000 shares authorized; 3,216 and 3,185 shares issued and outstanding as of December 31, 2024 and December 31, 2023, respectively	3	3
Additional paid-in capital	38,371	34,892
Accumulated other comprehensive loss	(670)	(143)
Retained earnings	35,209	27,882
Total stockholders' equity	72,913	62,634
Noncontrolling interests in subsidiaries	704	733
Total liabilities and equity	$ 122,070	$ 106,618

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Operations
(in millions, except per share data)

	Year Ended December 31,		
	2024	2023	2022
Revenues			
Automotive sales	$ 72,480	$ 78,509	$ 67,210
Automotive regulatory credits	2,763	1,790	1,776
Automotive leasing	1,827	2,120	2,476
Total automotive revenues	77,070	82,419	71,462
Energy generation and storage	10,086	6,035	3,909
Services and other	10,534	8,319	6,091
Total revenues	97,690	96,773	81,462
Cost of revenues			
Automotive sales	61,870	65,121	49,599
Automotive leasing	1,003	1,268	1,509
Total automotive cost of revenues	62,873	66,389	51,108
Energy generation and storage	7,446	4,894	3,621
Services and other	9,921	7,830	5,880
Total cost of revenues	80,240	79,113	60,609
Gross profit	17,450	17,660	20,853
Operating expenses			
Research and development	4,540	3,969	3,075
Selling, general and administrative	5,150	4,800	3,946
Restructuring and other	684	—	176
Total operating expenses	10,374	8,769	7,197
Income from operations	7,076	8,891	13,656
Interest income	1,569	1,066	297
Interest expense	(350)	(156)	(191)
Other income (expense), net	695	172	(43)
Income before income taxes	8,990	9,973	13,719
Provision for (benefit from) income taxes	1,837	(5,001)	1,132
Net income	7,153	14,974	12,587
Net income (loss) attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	62	(23)	31
Net income attributable to common stockholders	$ 7,091	$ 14,997	$ 12,556
Net income per share of common stock attributable to common stockholders			
Basic	$ 2.23	$ 4.73	$ 4.02
Diluted	$ 2.04	$ 4.30	$ 3.62
Weighted average shares used in computing net income per share of common stock			
Basic	3,197	3,174	3,130
Diluted	3,498	3,485	3,475

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Comprehensive Income
(in millions)

	Year Ended December 31,		
	2024	**2023**	**2022**
Net income	$ 7,153	$ 14,974	$ 12,587
Other comprehensive income (loss):			
Foreign currency translation adjustment	(539)	198	(392)
Unrealized net gain (loss) on investments, net of tax	12	16	(23)
Net loss realized and included in net income	—	4	—
Comprehensive income	6,626	15,192	12,172
Less: Comprehensive income (loss) attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	62	(23)	31
Comprehensive income attributable to common stockholders	$ 6,564	$ 15,215	$ 12,141

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Cash Flows
(in millions)

	Year Ended December 31,		
	2024	**2023**	**2022**
Cash Flows from Operating Activities			
Net income	$ 7,153	$ 14,974	$ 12,587
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation, amortization and impairment	5,368	4,667	3,747
Stock-based compensation	1,999	1,812	1,560
Inventory and purchase commitments write-downs	335	463	177
Foreign currency transaction net unrealized (gain) loss	(73)	(144)	81
Deferred income taxes	477	(6,349)	(196)
Non-cash interest and other operating activities	172	81	340
Digital assets (gain) loss, net	(589)	—	140
Changes in operating assets and liabilities:			
Accounts receivable	(1,083)	(586)	(1,124)
Inventory	937	(1,195)	(6,465)
Operating lease vehicles	(590)	(1,952)	(1,570)
Prepaid expenses and other assets	(3,273)	(2,652)	(3,713)
Accounts payable, accrued and other liabilities	3,588	2,605	8,029
Deferred revenue	502	1,532	1,131
Net cash provided by operating activities	14,923	13,256	14,724
Cash Flows from Investing Activities			
Purchases of property and equipment excluding finance leases, net of sales	(11,339)	(8,898)	(7,158)
Purchases of solar energy systems, net of sales	(3)	(1)	(5)
Proceeds from sales of digital assets	—	—	936
Purchase of intangible assets	—	—	(9)
Purchases of investments	(35,955)	(19,112)	(5,835)
Proceeds from maturities of investments	28,310	12,353	22
Proceeds from sales of investments	200	138	—
Receipt of government grants	—	—	76
Business combinations, net of cash acquired	—	(64)	—
Net cash used in investing activities	(18,787)	(15,584)	(11,973)
Cash Flows from Financing Activities			
Proceeds from issuances of debt	5,744	3,931	—
Repayments of debt	(2,500)	(1,351)	(3,364)
Proceeds from exercises of stock options and other stock issuances	1,241	700	541
Principal payments on finance leases	(381)	(464)	(502)
Debt issuance costs	(14)	(29)	—
Distributions paid to noncontrolling interests in subsidiaries	(104)	(144)	(157)
Payments for buy-outs of noncontrolling interests in subsidiaries	(133)	(54)	(45)
Net cash provided by (used in) financing activities	3,853	2,589	(3,527)
Effect of exchange rate changes on cash and cash equivalents and restricted cash	(141)	4	(444)
Net (decrease) increase in cash and cash equivalents and restricted cash	(152)	265	(1,220)
Cash and cash equivalents and restricted cash, beginning of period	17,189	16,924	18,144
Cash and cash equivalents and restricted cash, end of period	$ 17,037	$ 17,189	$ 16,924
Supplemental Non-Cash Investing and Financing Activities			
Acquisitions of property and equipment included in liabilities	$ 1,410	$ 2,272	$ 2,148
Supplemental Disclosures			
Cash paid during the period for interest	$ 277	$ 126	$ 152
Cash paid during the period for income taxes, net of refunds	$ 1,331	$ 1,119	$ 1,203

The accompanying notes are an integral part of these consolidated financial statements.

Revenue Recognition

Revenue by source

The following table disaggregates our revenue by major source (in millions):

	Year Ended December 31,		
	2024	2023	2022
Automotive sales	$ 72,480	$ 78,509	$ 67,210
Automotive regulatory credits	2,763	1,790	1,776
Energy generation and storage sales	9,564	5,515	3,376
Services and other	10,534	8,319	6,091
Total revenues from sales and services	95,341	94,133	78,453
Automotive leasing	1,827	2,120	2,476
Energy generation and storage leasing	522	520	533
Total revenues	$ 97,690	$ 96,773	$ 81,462

Automotive Segment

<u>Automotive Sales</u>

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, and specific other features and services that meet the definition of a performance obligation under ASC 606, including access to our FSD(Supervised) features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. We recognize revenue on automotive sales upon delivery to the customer, which is when the control of a vehicle transfers. Payments are typically received at the point control transfers or in accordance with payment terms customary to the business, except sales we finance for which payments are collected over the contractual loan term. We also recognize a sales return reserve based on historical experience plus consideration for expected future market values, when we offer resale value guarantees or similar buyback terms. Other features and services such as access to our internet connectivity, unlimited free Supercharging and over-the-air software updates are provisioned upon control transfer of a vehicle and recognized over time on a straight-line basis as we have a stand-ready obligation to deliver such services to the customer. Other limited free Supercharging incentives are recognized based on actual usage or expiration, whichever is earlier. We recognize revenue related to these other features and services over the performance period, which is generally the expected ownership life of the vehicle. Revenue related to FSD (Supervised) features is recognized when functionality is delivered to the customer and their ongoing maintenance is recognized over time. For our obligations related to automotive sales, we estimate standalone selling price by considering costs used to develop and deliver the service, third-party pricing of similar options and other information that may be available.

Any fees that are paid or payable by us to a customer’s lender when we arrange the financing are recognized upfront as an offset against automotive sales revenue. Costs to obtain a contract mainly relate to commissions for the sale of vehicles. As our contract costs related to automotive sales are typically fulfilled within one year, the costs to obtain a contract are expensed as incurred. Amounts billed to customers related to shipping and handling are classified as automotive sales revenue, and we have elected to recognize the cost for freight and shipping when control over vehicles, parts or accessories have transferred to the customer as an expense in cost of automotive sales revenue. Our policy is to exclude taxes collected from a customer from the transaction price of automotive contracts.

We offer resale value guarantees to our commercial banking partners in connection with certain vehicle leasing programs. Under these programs, we originate the lease with our end customer and immediately transfer the lease and the underlying vehicle to our commercial banking partner, with the transaction being accounted for as a sale under ASC 606.

Note 7 – Property, Plant and Equipment, Net

Our property, plant and equipment, net, consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Machinery, equipment, vehicles and office furniture	$ 18,339	$ 16,309
Land and buildings	10,677	9,498
AI infrastructure	5,152	1,510
Tooling	3,883	3,129
Leasehold improvements	3,688	3,136
Computer equipment, hardware and software	2,902	2,409
Construction in progress	6,783	5,791
	51,424	41,782
Less: Accumulated depreciation	(15,588)	(12,057)
Total	$ 35,836	$ 29,725

Construction in progress is primarily comprised of ongoing construction and expansion of our facilities, equipment and tooling related to the manufacturing of our products as well as AI-related assets which have not yet been placed in service. Completed assets are transferred to their respective asset classes and depreciation begins when an asset is ready for its intended use.

Depreciation expense during the years ended December 31, 2024, 2023 and 2022 was $4.12 billion, $3.33 billion and $2.42 billion, respectively.

Note 8 – Accrued Liabilities and Other

Our accrued liabilities and other current liabilities consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Accrued purchases (1)	$ 2,253	$ 2,721
Accrued warranty reserve, current portion	1,917	1,546
Payroll and related costs	1,532	1,325
Taxes payable (2)	1,367	1,204
Customer deposits	993	876
Operating lease liabilities, current portion	807	672
Sales return reserve, current portion	305	219
Other current liabilities	1,549	517
Total	$ 10,723	$ 9,080

(1) Accrued purchases primarily reflects receipts of goods and services for which we had not yet been invoiced. As we are invoiced for these goods and services, this balance will reduce and accounts payable will increase.

(2) Taxes payable primarily includes value added tax, income tax, sales tax, property tax and use tax payables.

Note 9 – Other Long-Term Liabilities

Our other long-term liabilities consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Operating lease liabilities	$ 4,603	$ 3,671
Accrued warranty reserve	4,799	3,606
Other non-current liabilities	1,093	876
Total other long-term liabilities	$ 10,495	$ 8,153

Note 10 – Debt

The following is a summary of our debt and finance leases as of December 31, 2024 (in millions):

	Net Carrying Value		Unpaid Principal Balance	Unused Committed Amount (1)	Contractual Interest Rates	Contractual Maturity Date
	Current	Long-Term				
Recourse debt:						
RCF Credit Agreement	—	—	—	5,000	Not applicable	January 2028
Other	4	3	7	—	4.70-5.75%	March 2025-January 2031
Total recourse debt	4	3	7	5,000		
Non-recourse debt:						
Automotive Asset-backed Notes	2,255	2,059	4,329	—	3.45-6.57%	September 2025-June 2035
China Working Capital Facility	—	2,740	2,740	—	1.92 %	April 2025 (2)
Energy Asset-backed Notes	54	434	493	—	4.80-6.25%	December 2025-June 2050
Cash Equity Debt	30	299	338	—	5.25-5.81%	July 2033-January 2035
Total non-recourse debt	2,339	5,532	7,900	—		
Total debt	2,343	5,535	$ 7,907	$ 5,000		
Finance leases	113	222				
Total debt and finance leases	$ 2,456	$ 5,757				

The following is a summary of our debt and finance leases as of December 31, 2023 (in millions):

	Net Carrying Value		Unpaid Principal Balance	Unused Committed Amount (1)	Contractual Interest Rates	Contractual Maturity Date
	Current	Long-Term				
Recourse debt:						
2024 Notes	$ 37	$ —	$ 37	$ —	2.00 %	May 2024
RCF Credit Agreement	—	—	—	5,000	Not applicable	January 2028
Other	—	7	7	28	4.70-5.75%	March 2025-January 2031
Total recourse debt	37	7	44	5,028		
Non-recourse debt:						
Automotive Asset-backed Notes	1,906	2,337	4,259	—	0.60-6.57%	July 2024-May 2031
Cash Equity Debt	28	330	367	—	5.25-5.81%	July 2033-January 2035
Energy Asset-backed Notes	4	8	13	—	4.80 %	December 2026
Total non-recourse debt	1,938	2,675	4,639	—		
Total debt	1,975	2,682	$ 4,683	$ 5,028		
Finance leases	398	175				
Total debt and finance leases	$ 2,373	$ 2,857				

(1) There are no restrictions on draw-down or use for general corporate purposes with respect to any available committed funds under our RCF Credit Agreement, except certain specified conditions prior to draw-down. Refer to the section below for the terms of the facility.

(2) The contractual maturity date of the China Working Capital Facility is April 2025, renewable until March 2026 at our discretion. As we have the intent and ability to refinance the loan on a long-term basis, we recorded it in Debt and finance leases, net of current portion in the consolidated balance sheets.

China Working Capital Facility

In April 2024, one of our subsidiaries entered into a loan agreement (the "China Working Capital Facility") with lenders in China for an unsecured revolving facility of up to RMB 20.00 billion to be used for certain production expenditures as well as repayment of certain finance facilities. Borrowed funds bear interest at a rate equal to the Loan Prime Rate published by the People's Bank of China minus 1.18%. The China Working Capital Facility is non-recourse to our assets.

Pledged Assets

As of December 31, 2024 and 2023, we had pledged or restricted $5.16 billion and $4.64 billion of our assets (consisting principally of operating lease vehicles, financing receivables, restricted cash, and equity interests in certain SPEs) as collateral for our outstanding debt.

Schedule of Principal Maturities of Debt

The future scheduled principal maturities of debt as of December 31, 2024 were as follows (in millions):

	Recourse debt	Non-recourse debt	Total
2025	$ 4	$ 2,349	$ 2,353
2026	—	4,116	4,116
2027	—	699	699
2028	—	243	243
2029	—	95	95
Thereafter	3	398	401
Total	$ 7	$ 7,900	$ 7,907

Note 11 – Leases

We have entered into various operating and finance lease agreements for certain of our offices, manufacturing and warehouse facilities, retail and service locations, data centers, equipment, vehicles, and solar energy systems, worldwide. We determine if an arrangement is a lease, or contains a lease, at inception and record the lease in our financial statements upon lease commencement, which is the date when the underlying asset is made available for use by the lessor.

We have lease agreements with lease and non-lease components, and have elected to utilize the practical expedient to account for lease and non-lease components together as a single combined lease component, from both a lessee and lessor perspective with the exception of direct sales-type leases and production equipment classes embedded in supply agreements. From a lessor perspective, the timing and pattern of transfer are the same for the non-lease components and associated lease component and, the lease component, if accounted for separately, would be classified as an operating lease.

We have elected not to present short-term leases on the consolidated balance sheet as these leases have a lease term of 12 months or less at lease inception and do not contain purchase options or renewal terms that we are reasonably certain to exercise. All other lease assets and lease liabilities are recognized based on the present value of lease payments over the lease term at commencement date. Because most of our leases do not provide an implicit rate of return, we used our incremental borrowing rate based on the information available at lease commencement date in determining the present value of lease payments.

Our leases, where we are the lessee, often include options to extend the lease term for up to 10 years. Some of our leases also include options to terminate the lease prior to the end of the agreed upon lease term. For purposes of calculating lease liabilities, lease terms include options to extend or terminate the lease when it is reasonably certain that we will exercise such options.

Lease expense for operating leases is recognized on a straight-line basis over the lease term as cost of revenues or operating expenses depending on the nature of the leased asset. Certain operating leases provide for annual increases to lease payments based on an index or rate. We calculate the present value of future lease payments based on the index or rate at the lease commencement date for new leases. Differences between the calculated lease payment and actual payment are expensed as incurred. Amortization of finance lease assets is recognized over the lease term as cost of revenues or operating expenses depending on the nature of the leased asset. Interest expense on finance lease liabilities is recognized over the lease term within Interest expense in the consolidated statements of operations.

The balances for the operating and finance leases where we are the lessee are presented as follows (in millions) within our consolidated balance sheets:

	December 31, 2024	December 31, 2023
Operating leases:		
Operating lease right-of-use assets	$ 5,160	$ 4,180
Accrued liabilities and other	$ 807	$ 672
Other long-term liabilities	4,603	3,671
Total operating lease liabilities	$ 5,410	$ 4,343
Finance leases:		
Solar energy systems, net	$ 21	$ 23
Property, plant and equipment, net	350	601
Total finance lease assets	$ 371	$ 624
Current portion of long-term debt and finance leases	$ 113	$ 398
Long-term debt and finance leases, net of current portion	222	175
Total finance lease liabilities	$ 335	$ 573

The components of lease expense are as follows (in millions) within our consolidated statements of operations:

	Year Ended December 31,		
	2024	2023	2022
Operating lease expense:			
Operating lease expense (1)	$ 1,500	$ 1,153	$ 798
Finance lease expense:			
Amortization of leased assets	$ 409	$ 506	$ 493
Interest on lease liabilities	21	45	72
Total finance lease expense	$ 430	$ 551	$ 565
Total lease expense	$ 1,930	$ 1,704	$ 1,363

(1) Includes short-term leases and variable lease costs, which are immaterial.

Other information related to leases where we are the lessee is as follows:

	December 31, 2024	December 31, 2023
Weighted-average remaining lease term:		
Operating leases	7.9 years	7.4 years
Finance leases	4.4 years	2.3 years
Weighted-average discount rate:		
Operating leases	5.3 %	5.6 %
Finance leases	4.7 %	5.5 %

Supplemental cash flow information related to leases where we are the lessee is as follows (in millions):

	Year Ended December 31,		
	2024	2023	2022
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash outflows from operating leases	$ 1,450	$ 1,084	$ 754
Operating cash outflows from finance leases (interest payments)	$ 21	$ 47	$ 75
Leased assets obtained in exchange for finance lease liabilities	$ 144	$ 10	$ 58
Leased assets obtained in exchange for operating lease liabilities	$ 1,808	$ 2,170	$ 1,059

As of December 31, 2024, the maturities of our operating and finance lease liabilities (excluding short-term leases) are as follows (in millions):

	Operating Leases	Finance Leases
2025	$ 1,066	$ 122
2026	958	84
2027	843	69
2028	742	27
2029	642	22
Thereafter	2,428	45
Total minimum lease payments	6,679	369
Less: Interest	1,269	34
Present value of lease obligations	5,410	335
Less: Current portion	807	113
Long-term portion of lease obligations	$ 4,603	$ 222

As of December 31, 2024, we have excluded from the table above additional operating leases that have not yet commenced with aggregate rent payments of $1.10 billion. These operating leases will commence between fiscal year 2025 and 2026 with lease terms of 2 years to 20 years.

Operating Lease and Sales-type Lease Receivables

We are the lessor of certain vehicle and solar energy system arrangements as described in Note 2, *Summary of Significant Accounting Policies*. As of December 31, 2024, maturities of our operating lease and sales-type lease receivables from customers for each of the next five years and thereafter were as follows (in millions):

	Operating Leases	Sales-type Leases
2025	$ 1,265	$ 177
2026	777	185
2027	454	93
2028	240	27
2029	200	2
Thereafter	1,290	—
Gross lease receivables	$ 4,226	$ 484

The above table does not include vehicle sales to customers or leasing partners with a resale value guarantee as the cash payments were received upfront. For our solar PPA arrangements, customers are charged solely based on actual power produced by the installed solar energy system at a predefined rate per kilowatt-hour of power produced. The future payments from such arrangements are not included in the above table as they are a function of the power generated by the related solar energy systems in the future.

Net Investment in Sales-type Leases

Net investment in sales-type leases, which is the sum of the present value of the future contractual lease payments, is presented on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Lease receivables relating to sales-type leases are presented on the consolidated balance sheets as follows (in millions):

	December 31, 2024	December 31, 2023
Gross lease receivables	$ 484	$ 780
Unearned interest income	(38)	(78)
Allowance for expected credit losses	(6)	(6)
Net investment in sales-type leases	$ 440	$ 696
Reported as:		
Prepaid expenses and other current assets	$ 152	$ 189
Other non-current assets	288	507
Net investment in sales-type leases	$ 440	$ 696

Note 12 – Equity Incentive Plans

In June 2019, we adopted the 2019 Equity Incentive Plan (the “2019 Plan”). The 2019 Plan provides for the grant of stock options, restricted stock, RSUs, stock appreciation rights, performance units and performance shares to our employees, directors and consultants. Stock options granted under the 2019 Plan may be either incentive stock options or nonstatutory stock options. Incentive stock options may only be granted to our employees. Nonstatutory stock options may be granted to our employees, directors and consultants. Generally, our stock options and RSUs vest over four years and our stock options are exercisable over a maximum period of 10 years from their grant dates. Vesting typically terminates when the employment or consulting relationship ends.

As of December 31, 2024, 113.0 million shares were reserved and available for issuance under the 2019 Plan.

As of December 31, 2024, we maintained valuation allowances of $1.22 billion for deferred tax assets that are not more likely than not to be realized, which primarily included our California deferred tax assets, U.S. foreign tax credits and certain foreign operating losses. The valuation allowance on our net deferred tax assets increased by $332 million during the year ended December 31, 2024, and decreased by $6.46 billion and $1.73 billion during the years ended December 31, 2023 and 2022, respectively. The valuation allowance increase during the year ended December 31, 2024 was primarily due to the changes of our California deferred tax assets, U.S. foreign tax credits and certain foreign operating losses. The change in valuation allowance during the year ended December 31, 2023 was primarily due to the release of our valuation allowance with respect to our U.S. federal and certain state deferred tax assets. In the fourth quarter of 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which was objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that most of our U.S. federal and certain state deferred tax assets are realizable and released the valuation allowance on these deferred tax assets. The valuation allowance change during the years ended December 31, 2022 was primarily due to changes in our U.S. deferred tax assets and liabilities. Our deferred tax assets without a valuation allowance are more likely than not to be realized given the expectation of future earnings in the respective jurisdictions.

As of December 31, 2024, we had $4.34 billion of federal and $8.59 billion of state net operating loss carry-forwards available to offset future taxable income, an immaterial amount of which, if not utilized, will begin to expire in 2026 for federal and 2025 for state purposes. Federal and state laws can impose substantial restrictions on the utilization of net operating loss and tax credit carry-forwards in the event of an “ownership change,” as defined in Section 382 of the Internal Revenue Code. We have determined that no significant limitation would be placed on the utilization of our net operating loss and tax credit carry-forwards due to prior ownership changes or expirations.

As of December 31, 2024, we had federal research and development tax credits of $1.48 billion, federal renewable energy tax credits of $1.01 billion, and state research and development tax credits of $1.06 billion. Most of our state research and development tax credits were in the state of California. If not utilized, some of the federal tax credits may expire in various amounts beginning in 2035. However, California research and development tax credits can be carried forward indefinitely.

As of December 31, 2024, we intend to indefinitely reinvest our foreign earnings and cash unless such repatriation results in no or minimal tax costs. We have recorded the taxes associated with the foreign earnings we intend to repatriate in the future. For the earnings we intend to indefinitely reinvest, no deferred tax liabilities for foreign withholding or other taxes have been recorded. The estimated amount of such unrecognized withholding tax liability associated with the indefinitely reinvested earnings is approximately $309 million.

Uncertain Tax Positions

The changes to our gross unrecognized tax benefits were as follows (in millions):

December 31, 2021	$ 531
Increases in balances related to prior year tax positions	136
Decreases in balances related to prior year tax positions	(12)
Increases in balances related to current year tax positions	222
Decreases in balances related to expiration of the statute of limitations	(7)
December 31, 2022	870
Increases in balances related to prior year tax positions	59
Decreases in balances related to settlement with tax authorities	(6)
Increases in balances related to current year tax positions	255
Decreases in balances related to expiration of the statute of limitations	(4)
December 31, 2023	1,174
Increases in balances related to prior year tax positions	51
Decreases in balances related to prior year tax positions	(27)
Increases in balances related to current year tax positions	227
Decreases in balances related to settlement with tax authorities	(4)
Decreases in balances related to expiration of the statute of limitations	(4)
December 31, 2024	$ 1,417

Note 17 – Segment Reporting and Information about Geographic Areas

Our Chief Executive Officer, as the CODM, organizes our company, manages resource allocations and measures performance among two operating and reportable segments: (i) automotive and (ii) energy generation and storage. The automotive segment includes the design, development, manufacturing, sales and leasing of electric vehicles as well as sales of automotive regulatory credits. Additionally, the automotive segment is also comprised of services and other, which includes sales of used vehicles, non-warranty maintenance services and collision, part sales, paid Supercharging, insurance services revenue and retail merchandise sales. The energy generation and storage segment includes the design, manufacture, installation, sales and leasing of solar energy generation and energy storage products and related services and sales of solar energy systems incentives.

Our CODM does not evaluate operating segments using asset or liability information. The CODM uses gross profit to allocate operating and capital resources and assesses performance of each segment by comparing actual gross profit results to historical results and previously forecasted financial information. The following table presents revenues, cost of revenues and gross profit by reportable segment (in millions):

	Year Ended December 31,		
	2024	2023	2022
Automotive segment			
Revenues	$ 87,604	$ 90,738	$ 77,553
Cost of revenues (1)	$ 72,794	$ 74,219	$ 56,988
Gross profit	$ 14,810	$ 16,519	$ 20,565
Energy generation and storage segment			
Revenues	$ 10,086	$ 6,035	$ 3,909
Cost of revenues (2)	$ 7,446	$ 4,894	$ 3,621
Gross profit	$ 2,640	$ 1,141	$ 288

(1) Depreciation and amortization included in Cost of revenues for the automotive segment for the years ended December 31, 2024, 2023 and 2022 was $3.68 billion, $3.45 billion and $2.81 billion, respectively.

(2) Depreciation and amortization included in Cost of revenues for the energy generation and storage segment for the years ended December 31, 2024, 2023 and 2022 was $377 million, $343 million and $331 million, respectively.

The following table presents revenues by geographic area based on the sales location of our products (in millions):

	Year Ended December 31,		
	2024	2023	2022
United States	$ 47,725	$ 45,235	$ 40,553
China	20,944	21,745	18,145
Other international	29,021	29,793	22,764
Total	$ 97,690	$ 96,773	$ 81,462

The following table presents long-lived assets by geographic area (in millions):

	December 31, 2024	December 31, 2023
United States	$ 32,461	$ 26,629
Germany	4,175	4,258
Other international	4,124	4,067
Total	$ 40,760	$ 34,954

The following table presents inventory by reportable segment (in millions):

	December 31, 2024	December 31, 2023
Automotive	$ 9,988	$ 11,139
Energy generation and storage	2,029	2,487
Total	$ 12,017	$ 13,626

Note 18 – Restructuring and Other

In the second quarter of 2024, we initiated and substantially completed certain restructuring actions to reduce costs and improve efficiency. As a result, we recognized $583 million of employee termination expenses in Restructuring and other in our consolidated income statement. These expenses were substantially paid with an immaterial accrual remaining in Accrued liabilities and other in our consolidated balance sheet as of December 31, 2024.

During the year ended December 31, 2022, we recorded an impairment loss of $204 million as well as realized gains of $64 million in connection with converting our holdings of digital assets into fiat currency. We also recorded other expenses of $36 million during the second quarter of the year ended December 31, 2022, related to employee terminations.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our Chief Executive Officer and our Chief Financial Officer, evaluated the effectiveness of our disclosure controls and procedures pursuant to Rule 13a-15 under the Securities Exchange Act of 1934, as amended (the “Exchange Act”). In designing and evaluating the disclosure controls and procedures, our management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving the desired control objectives. In addition, the design of disclosure controls and procedures must reflect the fact that there are resource constraints and that our management is required to apply its judgment in evaluating the benefits of possible controls and procedures relative to their costs.

Based on this evaluation, our Chief Executive Officer and our Chief Financial Officer concluded that, as of December 31, 2024, our disclosure controls and procedures were designed at a reasonable assurance level and were effective to provide reasonable assurance that the information we are required to disclose in reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC rules and forms, and that such information is accumulated and communicated to our management, including our Chief Executive Officer and our Chief Financial Officer, as appropriate, to allow timely decisions regarding required disclosures.

Management’s Report on Internal Control over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is a process designed by, or under the supervision of, our Chief Executive Officer and Chief Financial Officer to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles and includes those policies and procedures that (1) pertain to the maintenance of records that in reasonable detail accurately and fairly reflect the transactions and dispositions of our assets; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that our receipts and expenditures are being made only in accordance with authorizations of our management and directors and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of our assets that could have a material effect on the financial statements.